

BeatMaker — User Manual

Version: 1.02 | Script type: Praat script (.praat)

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Repository: <https://github.com/leonidasjr/prosody-apps>

Reference paper: Silva Jr. (2023) — RENOTE, v. 21, n. 1. DOI: <https://doi.org/10.22456/1679-1916.134363>

1. What BeatMaker does

BeatMaker is a multilingual, prosody-based script for Praat (Boersma & Weenink, 2021) that compares two speech recordings by extracting and overlaying their fundamental frequency (F0) contours and synthesizing beat (rhythmic pulse) signals from each voice.

Given a reference audio file (e.g., a native speaker — labelled L1) and a comparison audio file (e.g., an L2 learner — labelled L2), together with a plain-text transcription of the reference utterance, BeatMaker:

- Performs automatic forced alignment at the phrase level (intensity-based) and at the word level (Praat's built-in aligner).
- Extracts the F0 (pitch) contour of each speaker within sex-appropriate or user-defined frequency ranges.
- Synthesizes a beat signal (a stereo object combining rhythmic glottal pulses with the original waveform) and a pitch humming signal for each voice.
- Renders a multicolored overlay plot of both F0 contours with the aligned TextGrid and saves it as a 600-dpi JPEG image.

The beat in BeatMaker's name refers to the rhythmic pulsing pattern extracted from the glottal source via Praat's pulses method — making the timing and stress contour of the utterance audible as a percussion-like signal, analogous to the beat in music. This metaphor is pedagogically powerful: learners can literally hear where the stressed beats fall in native speech and compare them to their own production.

2. Theoretical background

2.1 L2 prosody in language teaching

Pronunciation accuracy in a second language (L2) requires mastering both segmental features (individual consonants and vowels) and suprasegmental/prosodic features of speech: stress, rhythm, intonation, tempo, and voice quality (Lengeris, 2012; Fletcher, 2010; Jackson & O'Brien, 2011). Inadequate production of L2 prosody leads to misunderstandings at the word, sentence, and discourse levels, and significantly reduces intelligibility (Derwing & Rossiter, 2003; Levis, 2018).

For learners whose L1 is a syllable-timed language (such as Brazilian Portuguese), producing the stress-timed rhythm of English is a particular challenge. Celce-Murcia et al. (2010) demonstrate that training on stress-based rhythm helps EFL learners reduce their foreign accent and enhance oral communication fluency.

2.2 Visual and auditory feedback in CAPT

Computer-Assisted Pronunciation Training (CAPT) systems that provide visual feedback on prosodic features have been shown to accelerate L2 prosody acquisition (Chun et al., 2008; Hardison, 2004; Derwing & Munro, 2015). Pyshkin et al. (2019) developed an acoustically-based CAPT system that graphically compares a learner's pitch pattern with a native-speaker model using two pitch curves — a design directly aligned with BeatMaker's approach. BeatMaker extends this paradigm by adding an auditory beat dimension.

2.3 Prosodic levels targeted

BeatMaker is optimized for speech chunks at three prosodic levels (Nespor & Vogel, 2007 [1986]):

Level	Symbol	Example
Prosodic word	ω	[girl] ω , [danced] ω
Prosodic phrase	ϕ	[the girl] ϕ [danced well] ϕ
Intonational phrase	I	[[the girl danced well] ϕ [because she had practiced] ϕ]I

► For best results, provide utterances at the prosodic phrase or intonational phrase level.

2.4 F0 extraction and beat synthesis in Praat

Praat extracts F0 using an autocorrelation-based pitch estimation algorithm (Boersma, 1993). The F0 range is set according to the speaker's sex:

Speaker sex	F0 range	Typical use
Female	90–350 Hz	Female adult speakers
Male	75–200 Hz	Male adult speakers
Customize	User-defined	Children, atypical voices, clinical assessment

The beat signal is derived by: (1) converting the F0 object to a pulses signal (one pulse per glottal cycle); (2) converting the F0 object to a sine wave at 44,100 Hz (amplitude $\times$ 4 for audibility); and (3) combining sine and pulses into a stereo Beat object, then combining that with the original voice into a Voice & Beat stereo object.

3. Software requirements

Requirement	Details
Praat	Version $\geq$ 6.0 (free: https://www.fon.hum.uva.nl/praat/)
Operating system	Windows, macOS, or Linux
Internet connection	Not required; may be needed for word aligner on first use
Audio files	.wav format, mono channel preferred

Requirement	Details
Storage	< 5 MB for the full repository

4. Preparing your files

4.1 Audio files

- Both audio files must be .wav format. Convert stereo to mono in Praat: select Sound → Convert → Convert to mono.
- Place both .wav files and the script in the same folder.
- Mandatory file naming convention:

Voice role	Required prefix(es)	Example
Reference (L1 / native / target)	L1, NS, RV, or V1	L1_speech.wav
Comparison (L2 / learner / questioned)	L2, FS, CV, or V2	L2_speech.wav

4.2 Transcription file

Create a plain-text file (.txt, UTF-8 encoding) containing the orthographic transcription of the reference utterance. The .txt file must share the exact base name of the reference .wav file:

Reference audio: L1_speech.wav → Transcription: L1_speech.txt

For the example included in this repository, the .txt file contains:

Nobody talks to Paul because he looks mad

4.3 Folder structure

```
working_folder/
├── BeatMaker_v1_02.praat
├── L1_speech.wav
├── L2_speech.wav
└── L1_speech.txt
```

5. Step-by-step usage

1. Open Praat on your computer.
2. In the Praat Objects window, go to Praat menu → Open Praat script... and select BeatMaker_v1_02.praat.
3. In the script editor, click Run → Run (or Ctrl+R / Cmd+R). A form dialog will appear.
4. Fill in the form parameters (see Section 6) and click OK.
5. Praat will process the files. The Objects window will populate with TextGrid, Pitch, and Sound objects.

6. When the Save plots? dialog appears, click Continue to save the figure as BeatMaker_plot.jpg.
 7. A confirmation message appears in the Info window. You can now re-run with different parameters.
► After the script completes, select any *_Beat_and_Voice* object and click Play to hear the voice and rhythmic beat simultaneously through stereo headphones — the main pedagogical listening activity.
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6. Form fields explained

Field 1 — Intensity threshold (dB)

Default: 45.0 dB. This threshold controls the intensity floor used for automatic phrase-boundary detection. Frames above this value are treated as speech.

- Increase (e.g., 55–60 dB) if background noise is being detected as speech.
- Decrease (e.g., 35–40 dB) if soft-spoken passages are not being included.

Field 2 — Reference sound

Default: L1_speech. Enter the base name (without extension) of your reference .wav file. The script will look for <name>.wav and <name>.txt in the working folder.

Field 3 — Language

Default: English (America). Selects the language model for Praat's word-level forced aligner.

Field 4 — Speaker sex

Default: Male. Determines the F0 frequency range for pitch extraction (Female: 90–350 Hz; Male: 75–200 Hz; Customize: user-defined).

Field 4.1 — F0 thresholds (Customize only)

Active only when Speaker sex = Customize. Suggested values by population:

Population	Lower (Hz)	Upper (Hz)
Children (5–12 y.)	150	500
Elderly male	60	180
Elderly female	80	300
Transgender (MTF)	120	350

Field 5 — Pitch contour colors

L1 default: Red; L2 default: Blue. Choose from Black, Red, Blue, or Green. Use contrasting colors for classroom readability.

7. Script walkthrough — block by block

Block 1 — Header and citation (lines 1–25)

Defines the script's identity, authorship, purpose, and citation information as Praat comments. Documents the three application domains and strongly recommends two audio files per folder.

Block 2 — Form dialog (lines 27–97)

Declares the interactive form: intensity threshold, reference sound name, language option menu (9 languages), speaker sex (3 buttons), custom F0 bounds (integers), and color selectors for L1 and L2 contours.

Block 3 — Canvas and objects cleanup (lines 101–117)

Clears Praat's Picture window (Erase all) for a blank canvas. Checks for selected objects in the Objects window and removes them all to prevent conflicts with previous runs.

Block 4 — Reference audio loading and phrase-level TextGrid (lines 121–196)

Reads the reference .wav and .txt files. Opens the Sound in the editor to configure intensity display settings, then closes it. Creates an empty TextGrid with a sentence tier. Converts the Sound to an Intensity object and iterates through all frames: any frame above the threshold triggers a boundary insertion in the TextGrid. After the loop, collapses all internal boundaries into a single utterance interval and writes the transcription text into it.

Block 5 — Word-level forced alignment (lines 199–235)

For any valid language selection (1–9), selects the reference Sound and TextGrid together, opens the TextGrid editor, runs Alignment settings with the selected language, and executes Align interval for word-level forced alignment. Tier names are set to Text (phrase) and Words (word). All objects except the TextGrid are removed. The Words tier is duplicated to position 1 and the copy at position 3 is deleted, producing a clean two-tier TextGrid.

Block 6 — Main processing loop (lines 247–654)

Creates a list of all .wav files in the folder. Iterates over each file, reads the Sound, and extracts the first two characters of the filename as the speaker label (language_chr\$: L1, L2, NS, FS, etc.).

F0 and beat extraction (lines 269–347): Calls To Pitch with the appropriate F0 range. Converts the Pitch to pulses (one per glottal cycle) and to a sine wave at 44,100 Hz (amplitude $\times 4$). Combines sine and pulses into a stereo Beat object, converts to mono, combines with the original voice into the Voice & Beat stereo object. Removes temporary objects. Converts the Pitch to a hum sound (L^*_{Pitch}).

Plot procedures (lines 357–651): Sets font (Times, 14 pt). Routes reference files (L1/NS/RV/V1) through the reference branch, comparison files (L2/FS/CV/V2) through the comparison branch. For each, smooths the Pitch object (bandwidth: 2 Hz), sets the user-chosen color, and draws the F0 contour in the main viewport (0–10 $\times$ 0–4.5 inches). Calls procedure plot_f0_YX_axes (shared Y/X axis labels and tick marks, reference voice only), procedure plot_voice_and_beat_L1 or plot_voice_and_beat_L2 (waveform and beat in sub-viewports). Overlays the TextGrid in the F0 panel. Draws color-coded legend boxes in the top-right corner.

Block 7 — Plot saving and cleanup (lines 657–669)

Pauses with the Save plots? dialog. On confirmation, saves the Picture window as BeatMaker_plot.jpg (600 dpi) in the working folder. Writes a confirmation to the Info window. Removes the Strings object to complete cleanup.

8. Output objects and files

Object / File	Type	Description
L1_Beat_and_Voice	Sound (stereo)	Reference voice (upper ch.) + beat pulses (lower ch.). Play through stereo headphones.
L2_Beat_and_Voice	Sound (stereo)	Comparison voice (upper ch.) + beat pulses (lower ch.). Play through stereo headphones.
L1_Pitch	Sound (mono)	Humming signal sonifying the reference speaker's F0 contour over time.
L2_Pitch	Sound (mono)	Humming signal sonifying the comparison speaker's F0 contour over time.
TextGrid (reference)	TextGrid	Two tiers: Text (phrase-level interval with transcription) and Words (word-level forced alignment).
BeatMaker_plot.jpg	JPEG, 600 dpi	Overlay F0 contour plot with TextGrid alignment and Voice & Beat panels for both speakers.

Reading the plot

The BeatMaker_plot.jpg output contains three visual areas:

8. F0 contour panel (upper, full width): Both pitch contours overlaid in their respective colors, with the aligned TextGrid superimposed at mid-height. Y-axis = frequency in Hz; X-axis = time in seconds. Legend box (top right) identifies each speaker by color.
9. L1 Voice & Beat panel (lower left): Waveform of the reference voice (upper half) and rhythmic beat pulses (lower half), displayed as a stereo oscillogram. Makes the timing structure of native speech visible.
10. L2 Voice & Beat panel (lower right): Same display for the comparison voice.

9. Pedagogical and research applications

In the L2 pronunciation classroom

Suggested workflow:

11. Record a brief utterance from a native speaker as the reference file (L1_...wav).
12. Record the same utterance from each learner and save as L2_...wav.
13. Run BeatMaker for each learner pair, projecting the resulting plot for the class.
14. Guide learners to compare the F0 contours visually (rising/falling patterns, peak placement) and listen to the beat signals to identify rhythmic discrepancies.
15. Repeat after practice to track improvement over time.

► *Research note: A pretest-posttest experiment with 10 Brazilian EFL learners vs. 8 native North American English speakers showed a significant improvement in intonational aspects after*

BeatMaker training (convergence toward the native intonation pattern), with no significant effect on rhythmic parameters (Silva Jr., 2023).

In forensic phonetics

BeatMaker supports speaker comparison tasks by providing a visual and auditory overlay of F0 contours from a questioned recording and a reference recording. Use the V1/V2 or RV/CV naming convention for these cases.

In clinical voice assessment

Compare a patient's voice at different time points (e.g., before and after therapy) or contrast a pathological voice against a normative model. Use the Customize speaker sex option to set appropriate F0 ranges for clinical populations.

10. Troubleshooting

Symptom	Likely cause	Solution
WARNING! Your files must be tagged...	Filename prefix not recognized	Rename files: first two chars must be L1/L2/NS/FS/RV/CV/V1/V2
No word boundaries in TextGrid	Language model unavailable; no internet	Connect to the internet and re-run; or accept phrase-level alignment
F0 contour mostly blank or sparse	F0 range does not match the speaker	Use Customize and set a wider or different F0 range
Alignment misses utterance start/end	Threshold too high	Lower Intensity_threshold_(dB) to 35–40 dB
Plot saves as all-white or empty	Picture window cleared during the pause	Re-run; do not interact with Picture window during processing
File not found error on .txt file	.txt name does not match .wav base name	Verify both files share the exact same base name (no extension)
Stereo audio causes errors	Input file is stereo	Convert to mono: select Sound → Convert → Convert to mono

11. How to cite

Script:

```
SILVA JR., Leônidas. (2023–2025). BeatMaker (version 1.02).  
[Computer program for Praat]. Available at:  
https://github.com/leonidasjr/VowelCode
```

Validation paper:

SILVA JR., Leônidas José. BeatMaker: a computational system for foreign language pronunciation teaching based on speech prosody.

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v. 21, n. 1, p. 341-350, jul. 2023.

DOI: <https://doi.org/10.22456/1679-1916.134363>

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